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## Elon Musk's Big Bets

On April 2, 2017, Elon Musk announced that Tesla produced 25,000 cars in the first quarter, setting a new company record.<sup>1</sup> Within two weeks, Tesla's market value zoomed past Ford Motor Company (which manufactured more than 1.5 million cars per quarter) and General Motors (which made more than 2.5 million cars per quarter) to become the most valuable U.S. car manufacturer.<sup>2</sup> Elon Musk used the occasion to needle his naysayers, writing on Twitter: "Stormy weather in Shortville."<sup>3</sup>

By most measures, Elon Musk was on a roll: Beyond promising to scale Tesla by a factor of 10 between 2015 and 2018, Musk was vowing to deliver fully autonomous driving and an automated car-sharing service.<sup>4</sup> To support his aggressive plans, Tesla was constructing a giant battery factory in the desert outside Reno, Nevada. Everything about the factory was massive: when complete, it would be the largest building in the world.<sup>5</sup> This so-called "Gigafactory" would require \$5 billion in investment and would ultimately produce 105 gigawatt hours (GWh) of battery cells per year, more than the entire capacity of the world's li-ion (lithium ion) production in 2015.<sup>6</sup>

Musk also wanted to turn Tesla into a renewable energy company. In the fall of 2016, Musk made another big bet by orchestrating Tesla's acquisition of the rooftop solar installer SolarCity, where Musk was the chairman and largest shareholder. While the automotive and solar businesses were starkly different, Musk insisted that they were part of Tesla's goal of "accelerating the advent of sustainable energy."<sup>7</sup> The Gigafactory plus SolarCity would transform Tesla from an automaker into what Musk called the "world's only vertically integrated energy company offering end-to-end clean energy products to our customers."<sup>8</sup>

In his spare time, Musk was also the CEO of the commercial space flight company SpaceX. Musk insisted that SpaceX would begin transporting people to Mars by 2025, achieving his stated goal of making humanity a "multiplanetary species."<sup>9</sup> Transforming transportation, pushing forward renewable energy generation and storage, and colonizing Mars: such was the scope of Musk's ambition. Little wonder that a former colleague said that "Elon thinks bigger than just about anyone else I've ever met."<sup>10</sup> Even though Tesla and SolarCity were unprofitable and burning cash, for Musk the question was always: what's next? For Musk's board and shareholders, the question was different: was Musk, as his biographer put it, "a being sent from the future to save mankind from itself or a slick businessman dragging foolish investors along on grand cash-burning bets?"<sup>11</sup>

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## Elon Musk's Ventures

Elon Musk was born and raised in South Africa, and made his way to Canada in 1989 at the age of 17. Staying with relatives and working odd jobs, Musk enrolled in Queen's University in Ontario and then transferred to the University of Pennsylvania, where he earned degrees in economics and physics. He began a PhD program in applied physics at Stanford, but dropped out in 1995 to start an online map and directory firm called Zip2 with his brother. They sold the company to Compaq for \$300 million in 1999. Musk used the money from the sale to launch another startup called X.com, an online bank. In 2000, X.com merged with Confinity, another startup, which ran a money transfer service called PayPal. The combined firm took the name PayPal in 2001 with Musk as the CEO and largest shareholder until the firm was sold to eBay for \$1.5 billion in 2002.

### *SpaceX*

Not content with founding and selling two successful startups, Musk founded Space Exploration Technologies, better known as SpaceX, in 2002. He told a reporter in 2003 that "I like to be involved in things that change the world. The Internet did, and space will probably be more responsible for changing the world than anything else."<sup>12</sup> Reaching Mars was the long-term goal, and Musk claimed in 2012 to be on track to get a manned spacecraft to Mars in 10–15 years. In the short term, however, SpaceX concentrated on the commercialization of near-Earth exploration.<sup>13</sup>

Musk invested some \$100 million of his own money in SpaceX and nearly lost it all when SpaceX struggled to achieve a successful test launch of its first rocket. By 2006, SpaceX had won contracts to take commercial and NASA satellites into space despite not yet launching a rocket. Its first three launch attempts ended in failure. By September 2008, SpaceX was on the brink of collapse. As a colleague recalled, "Everything hinged on that launch. Elon had lost all his money, but this was more than his fortune at stake—it was his credibility."<sup>14</sup> The launch succeeded, however. Then, in July 2009, SpaceX had its first successful paying mission.<sup>15</sup>

In May 2012, SpaceX became the first commercial firm to successfully dock a vehicle with the International Space Station (ISS), and later that year it delivered its first load of cargo to the ISS.<sup>16</sup> In late 2013, SpaceX carried out its first successful launch of a commercial satellite. SpaceX achieved another milestone in December 2015, when it successfully landed a rocket after launch. Musk saw reusable rockets as an essential step toward making spaceflight truly affordable. SpaceX executives estimated that reusing rockets could cut its launch prices by 30%, from around \$61 million to \$43 million per launch.<sup>17</sup> SpaceX advanced further toward this goal in March 2017, when it successfully launched and landed a previously flown Falcon 9 rocket.

Elon Musk was not the only high-profile technology entrepreneur to pursue ventures in space. Amazon founder and CEO Jeff Bezos, the second-richest person on Earth, founded spaceflight company Blue Origin, which directly competed with SpaceX. In November 2015, Blue Origin was the first company in history to successfully launch and land a rocket during a mission to space, proving to the world that reusable rocket technology could work.<sup>18</sup> As of March 2017, Blue Origin was focused on launching its first crewed flight into space and building heavy lift rockets to further compete with SpaceX. In order to ensure the company had enough money to accomplish its goals, Bezos pledged to sell \$1 billion of his Amazon stock each year to finance Blue Origin.<sup>19</sup>

Despite these successes, launching rockets into space remained a risky activity, and SpaceX had lost a few rockets and payloads over the years. On September 1, 2016, a SpaceX rocket exploded on the launch pad at Cape Canaveral. While no one was hurt, the loss was a particularly high-profile one

because the satellite on board was to be used as a part of Facebook's effort to bring Internet access to regions of Africa, the Middle East, and Europe. Such setbacks had a significant financial impact on SpaceX: in 2015, SpaceX's revenue declined 6% after multiple years of strong growth, and the company recorded a wide operating loss of \$260 million following multiple years of small but positive operating income (see **Exhibit 1**).

An early 2015 funding round valued SpaceX at approximately \$12 billion, making it one of the most valuable venture-backed private companies in the world.<sup>20</sup> Meanwhile, the revenue generated by SpaceX's success in near-Earth missions provided the capital to continue working toward the ultimate goal of regular space travel to Mars. While SpaceX sought to address Musk's dreams in the stars, he also had great ambitions for revolutionizing travel on and under the Earth's surface.

### *The Hyperloop, The Boring Company, and Neuralink*

In August 2013, Musk announced his idea for yet another revolutionary mode of transportation, dubbed the Hyperloop, which Musk claimed would be a faster and cheaper replacement for high-speed rail. Inspired by pneumatic tubes once used to shuttle documents around offices, it would transport passengers at speeds of more than 700 miles per hour in pods enclosed in underground steel tubes under near-vacuum conditions. Given his other obligations, Musk did not attempt to commercialize the idea, but published an open-source white paper describing the technology. A few startups picked up the idea and began developing the technology. One of them, a firm called Hyperloop One, held a demonstration in the Nevada desert in the spring of 2016, propelling a sled one-half mile down a test track at speeds of over 300 miles per hour. The startup had raised \$150 million for the venture through January 2017.<sup>21</sup> At the end of 2016, Musk tweeted the launch of a Hyperloop-like venture called The Boring Company, which would build underground tunnels to combat traffic. Days after the tweet, he bought the website BoringCompany.com and staffed a SpaceX engineer to oversee the new venture.<sup>22</sup> Finally, in the spring of 2017, Musk started yet another company, called Neuralink, which sought to merge the human brain with computers. Musk planned to be CEO of both new startups.

### *Tesla*

Although Elon Musk was the face of Tesla, he was not one of its founders. The company was started in 2003 by Silicon Valley engineers with the goal of producing a high-performance electric sports car. Musk joined the company in 2004 (two years after launching SpaceX) as its chairman and led its fundraising efforts, which netted \$7.5 million in its first round. Musk became CEO in 2008, by which time he had invested \$55 million of his own money. The company raised \$260 million in its 2010 IPO, the first American car company to go public since Ford in 1956.<sup>23</sup> At the end of 2016, Musk remained the largest shareholder in Tesla with a 20.8% ownership stake in the company. By March 2017, Tesla had raised over \$9.3 billion in financing (see **Exhibit 2a**).

Musk articulated a grand vision for Tesla and the broader electrical vehicle (EV) industry as the key to sustainable transportation, in the context of the looming disaster of climate change. As he put it in 2011, "[I]n order to change the infrastructure such that we avoid having some sort of catastrophic situation [a century from now], we must act now, because we're talking about changing what will probably be 2 billion cars. You don't just change that overnight. A whole industry has to be born."<sup>24</sup> Musk saw Tesla's role as bringing that industry into being, with the long-term goal of creating an affordable electric vehicle.

Because the cost of electric vehicle technology, particularly battery technology, did not permit the construction of an appealing mass-market electric car in the early 2000s, Tesla and Musk decided to enter the market at the high end and move down-market over time. Musk, tongue-in-cheek, revealed

Tesla's "secret plan" in 2006: "1. Build sports car; 2. Use that money to build an affordable car; 3. Use that money to build an even more affordable car."<sup>25</sup> True to the plan, Tesla's first production vehicle, released in 2008, was a high-performance and high-priced sports car called the Roadster. Tesla only manufactured 2,500 Roadsters, but it demonstrated that an electric car could deliver superior performance. The 300 horsepower Roadster went from 0 to 60 in 3.7 seconds, had a top speed of 125 mph, and sold for about \$110,000 in 2009. Tesla's next vehicle was a luxury performance sedan called the Model S that was released in 2012 and designed to compete with Mercedes, BMW, and Audi. It was priced starting at \$70,000, although optional features (such as a larger battery to provide longer range) could push the price well past \$100,000. While not by any means a truly "affordable" car, total Model S sales rose from under 5,000 in 2012 to 150,000 by September 2016.<sup>26</sup>

Initial reviews for the Model S were very positive. In 2013, *Consumer Reports* gave the first model its highest rating ever, a 99 out of 100, and *Motor Trend* named the Model S its Car of the Year. Two years later, *Consumer Reports* gave an all-wheel-drive version of the Model S a score of 103 out of 100 for its combination of power and efficiency, prompting it to rescale its scoring system to bring the Model S down to a perfect 100. Surprisingly, reliability problems with the Model S led *Consumer Reports* to revoke its recommendation in late 2015, citing "a worse-than-average overall problem rate" based on a survey of 1,400 Model S owners.<sup>27</sup>

Tesla's reputation suffered a further blow when three Tesla vehicles were damaged by battery fires in 2013, caused when the battery pack, which was installed in the car's undercarriage, was damaged by striking roadway debris. Although no one was hurt in the accidents and Tesla pointed out that fires happened at a far higher rate in gasoline-powered cars, the fires raised potentially damaging concerns about battery safety. Tesla's stock fell nearly 4% on the news, and the National Highway Traffic Safety Administration (NHTSA) opened an investigation. In response, Tesla announced in early 2014 that it would modify the Model S to raise its ground clearance at highway speeds and that it would reinforce the vehicle's underbody armor in order, in Musk's words, "to bring this risk down to virtually zero."<sup>28</sup> The NHTSA subsequently closed its investigation, saying that "a defect trend has not been identified."<sup>29</sup> Safety concerns were renewed, however, when a Model S caught fire while charging at a supercharger in Norway in January 2016, completely destroying the vehicle. Tesla, which ultimately traced the cause to a short circuit, insisted this was an isolated incident and pointed out that vehicles had been charged at supercharging stations 2.5 million times without incident. Norwegian officials concluded that it represented an isolated event.<sup>30</sup>

Tesla released its next vehicle, an SUV called the Model X, in late 2015, and it had sold and delivered about 37,000 of the vehicles by March 2017.<sup>31</sup> Like the Model S, it received rave reviews for its performance, but also faced quality issues in the early months after its release, including problems with the vehicle's unique falcon-wing rear doors and, more seriously, a faulty seat latch that in some cases allowed the rear seats to fold forward during a collision, which led to a recall of 2,700 vehicles in 2016.<sup>32</sup>

Shortly after the launch of the Model X, Tesla announced the availability of a package of self-driving features, which it called "Autopilot." The system used cameras, radar, GPS, and other sensors to provide semi-autonomous highway driving, keeping the car in its lane, changing lanes when necessary, and maintaining a safe following distance. Musk and Tesla described the system as a "public beta," recommending that drivers keep their hands on the wheel and remain alert and ready to take control at all times. It soon became clear that drivers were not heeding such warnings, posting videos of themselves reading or watching videos while letting the car drive itself.<sup>33</sup> In May 2016, a Tesla driver was killed while driving in Autopilot mode when neither the car's sensors nor the driver detected a tractor-trailer crossing the highway and the vehicle collided at full speed with the trailer. The Tesla driver was reportedly watching a movie at the time of the crash.<sup>34</sup> Tesla responded by releasing

upgrades to the Autopilot software designed to improve its radar processing system to better avoid forward collisions and implementing additional mechanisms to prevent drivers from ignoring warnings to keep their hands on the wheel, but concerns about the risks of introducing autonomous driving features remained.<sup>35</sup> In January 2017, the NHTSA ruled that Tesla was not at fault and that it was the driver's responsibility to see the truck.<sup>36</sup>

In March 2016, Tesla unveiled a prototype of its long-anticipated mass-market vehicle, a \$35,000 sedan called the Model 3, scheduled for release in late 2017. It began taking pre-orders the same day and received nearly 375,000 orders (requiring a \$1,000 refundable deposit) in the first month.<sup>37</sup> Partly on the strength of the response to the Model 3, Musk announced in May a new goal of producing 500,000 vehicles annually by 2018, two years earlier than the target Tesla had previously set. The target was greeted with considerable skepticism, since Tesla had only produced about 84,000 cars in 2016, and had a track record of missing production targets and release dates.<sup>38</sup>

Glowing reviews, growing sales, and ambitious production targets did not translate into profits. Tesla had operated at a loss every year of its history, and had lost a total of nearly \$2.7 billion by the end of 2016 (see **Exhibit 3a**). In 2015, Tesla lost an estimated \$4,000 for every vehicle it sold.<sup>39</sup> In addition to financial challenges, Tesla was suffering from a number of senior staff departures: over two dozen managers left the company in 2016 due to "long hours in the rush to high-volume production, mission creep, and a tense culture." Analysts expected that Tesla's rapid growth, as well as Musk's self-described style as a "nano-manager," would likely exacerbate high-level turnover.<sup>40</sup>

**The Gigafactory** Musk first floated the idea of building a battery factory in November 2013, telling analysts that the company was considering constructing a "giant" battery plant that would generate "something comparable to all li-ion production in the world in one factory." In its letter to shareholders announcing the Gigafactory, Tesla described it as "a fully integrated industrial complex. Processed ore from mines will enter by railcar on one side and finished battery packs will exit on the other."<sup>41</sup> The construction of the battery factory continued Tesla's pattern of vertical integration, in contrast to longstanding outsourcing trends in the auto industry. In the words of one analyst, Tesla was "not only the most vertically integrated tech company you've ever seen; it's also the most vertically integrated car company since Ford in the 1920s."<sup>42</sup> Integration extended to sales and recharging. Tesla sold directly to consumers online or through company-owned stores (in states where that was allowed), bypassing the franchise dealer networks that car companies had traditionally employed. In addition, by March 2017, Tesla had built over 800 supercharger stations, mostly in the U.S. and Western Europe (with a few in China, Japan, Canada, and Australia), which enabled drivers to rapidly recharge their vehicles free of charge on long-distance trips.<sup>43</sup>

In February 2014, Tesla estimated that the Nevada Gigafactory would cost approximately \$5 billion, with Tesla committing to directly invest \$2 billion and the remainder coming from partners in the project. In tandem with this announcement, Tesla raised \$2 billion from public markets in convertible debt, in part to fund its portion of the Gigafactory investment.<sup>44</sup> Tesla announced the site for the factory in September 2014, and construction began soon thereafter. Over the next two years, both the rationale for the factory and its scale expanded. Originally slated to supply batteries for Tesla's growing vehicle production, in early 2015 Tesla announced that nearly a third of the batteries manufactured at the Gigafactory would be dedicated to its new stationary energy storage business (see below). In early 2017, the uses of the Gigafactory expanded even further to include production of the Model 3 electric motors and gear boxes.<sup>45</sup> To meet these additional needs, Tesla had purchased nearly 2,000 additional acres at the factory's Nevada location, accelerated its construction schedule, and tripled the factory's planned production capacity from 35 GWh to 105 GWh by 2018, which would be larger than total global li-ion battery production in 2013.<sup>46</sup> Tesla continued its partnership with Panasonic in the construction and

operation of the factory. Panasonic agreed to invest \$1.5 to \$2 billion and would be responsible for the manufacture of the battery cells, including purchasing and maintaining the necessary equipment. Tesla would assemble the cells into battery modules and packs, which it was currently doing in its vehicle manufacturing plant in California using cells shipped from Panasonic's factories in Japan.<sup>47</sup>

Tesla began calling the acquired SolarCity factory based in Buffalo, New York, the "Gigafactory 2," which would produce photovoltaic (PV) cells to be used for solar panels and non-solar roof products. In similar fashion to the Nevada Gigafactory, the company partnered with Panasonic for the construction of the Buffalo factory, conditional on the completion of Tesla's acquisition of SolarCity. In exchange for Panasonic covering the capital costs of the Buffalo facility (approximately \$250 million), Tesla made a long-term purchase agreement for the PV cells produced at the factory.<sup>48</sup>

During Tesla's 2016 annual earnings call, the company announced that it would finalize locations for a third, fourth, and potentially fifth Gigafactory, indicating that Tesla and Musk had no intention of slowing down after completing the Gigafactories in Nevada and Buffalo.<sup>49</sup> These plans also highlighted Elon Musk's belief that global demand for batteries had no intention of slowing down either. In April 2017, he announced that he would likely build four more Gigafactories.

## A Battery Company?

The scale of the Gigafactory led at least one observer to ask in early 2015 if Tesla would consider changing from being primarily a car company to primarily a battery maker, to which Musk replied, "It's possible."<sup>50</sup> The li-ion batteries to be manufactured in the Gigafactory were derived from the same basic technology as batteries used in cell phones, tablets, laptops, and other consumer electronic devices. Although a wide variety of battery technologies had emerged since the invention of the battery in 1799, li-ion became the dominant energy storage technology after it was first commercialized in 1991. The worldwide market for li-ion batteries was projected to grow from \$21.3 billion in 2014 to over \$76 billion in 2020, driven by the continued growth of consumer electronics, the growth of electric vehicles, and the emerging market for energy storage for residential, commercial, and utility-scale applications. At the same time, demand for li-ion was shifting; consumer electronics was by far the largest application in 2013, but it was expected that large-scale energy storage applications and electric vehicles would become the largest sources of demand by 2020 (see **Exhibit 4**).

To meet anticipated growth in demand, li-ion battery manufacturing capacity was growing rapidly across the world. BYD, Samsung, and Daimler had announced plans to expand manufacturing capacity by 2020.<sup>51</sup> There was evidence of excess capacity in the industry by 2015, as ramp-up of battery production had exceeded the growth in demand generated by EVs. But some analysts predicted that demand would catch up to capacity over the next 5–10 years.<sup>52</sup> Nevertheless, batteries of sufficient capacity to provide an electric car with performance and range comparable to that of a traditional gasoline-powered car remained very costly, and their cost represented the major barrier to the development of a genuine mass-market electric car. In many cases the cost of the battery represented half the cost of the car (although Tesla CTO JB Straubel claimed that in the case of the Model S, the battery represented as little as one-fourth the total cost of the vehicle in 2015). Estimates of the actual production cost of li-ion batteries, typically measured in dollars per kilowatt hour (\$/kWh), varied considerably, but analysts estimated the industry average for battery packs at approximately \$350/kWh in 2016 (down from over \$1,000/kWh in 2007). Some analysts put Tesla's battery costs as low as \$150 to \$200/kWh, due in part to the shape of its battery cells and the energy-dense materials used in its batteries.<sup>53</sup> The U.S. Department of Energy in 2012 had set a goal of reducing EV battery

costs to \$125/kWh by 2022, roughly the cost at which electric vehicles would become price-competitive with gasoline-powered cars.<sup>54</sup>

### *Alternative Battery Technologies*

In addition to developments within existing li-ion technology, researchers were investigating new battery technologies and experimenting with different elements to develop a cheaper battery. Despite promising advances, many industry analysts and battery researchers insisted that any post-lithium battery future remained at least a decade away in 2016, and li-ion remained the best battery technology in existence. Fuel cells represented an alternative energy storage and delivery technology that served much the same range of applications as li-ion battery technology. Although fuel cells could use a variety of fuels, hydrogen was the most common, and hydrogen fuel cells had been adopted for a variety of applications by 2015. When used to power electric vehicles, fuel cells had several advantages over battery-powered electric cars, including superior range (often on par with gasoline-powered cars) and superior storage capacity. Some observers saw fuel cells as a better long-term solution to the world's energy storage needs compared to li-ion batteries. In the words of one executive in the fuel cell industry, "I think the Tesla factory producing hundreds of thousands more li-ion batteries is really short-sighted because those batteries are just never going to hold the amount of energy we need them to."<sup>55</sup> Although battery-powered electric cars remained the primary alternative to the traditional gasoline-powered car, electric vehicles powered by hydrogen fuel cells (FCEVs) were emerging as a competitor in the zero-emission vehicle category.

## **Electric Vehicles**

Electric vehicles had existed as far back as the late nineteenth century, but were soon surpassed in popularity by cars with gasoline-powered internal combustion engines. Efforts to develop electric cars faltered over the next century, until rising oil prices, improvements in battery technology, and growing concerns about climate change spurred renewed interest in the feasibility of electric cars at the beginning of the twenty-first century. In addition to environmental benefits, electric motors had certain advantages over internal combustion engines: they were much smaller, less expensive, and simpler. Electric motors did not require a transmission, nor did they require the lubrication and exhaust systems associated with internal combustion engines.

Governments actively encouraged the adoption of electric vehicles as part of their efforts to address pollution and climate change. The U.S. government introduced a federal tax credit of \$7,500 for EV purchases in 2013, and, as of 2015, 37 U.S. states offered various incentives to promote hybrid and electric vehicle adoption, including rebates or tax credits ranging from \$1,000 to \$6,000, HOV lane exemptions, emissions test exemptions, and utility rate reductions.<sup>56</sup>

Despite such encouragements, electric cars remained a tiny, albeit growing, fraction of the total market. Overall, electric vehicles represented 160,000 of the 17.55 million cars sold in the U.S. in 2015.<sup>57</sup> Globally, sales of electric vehicles (including both plug-in hybrids and pure EVs like the Tesla) had risen from some 190,000 in 2013 to nearly 540,000 in 2015, representing 0.6% of the total global automobile market.<sup>58</sup> Sales continued to rise in 2016, up 42% in the first quarter compared to the same period the previous year.<sup>59</sup> As of 2016, the Nissan Leaf was the leading seller, and it had taken three years to sell its first 100,000 vehicles, while BMW delivered just over 24,000 of its i3 compact electric vehicle in the first half of 2015.<sup>60</sup>

In a bid to spur the growth of the industry, Musk announced in June 2014 that Tesla would "not initiate patent lawsuits against anyone who, in good faith, wants to use our technology." Musk claimed

that “Tesla, other companies making electric cars, and the world would all benefit from a common, rapidly-evolving technology platform.”<sup>61</sup> Other automakers followed suit. Toyota made a similar move the following January, opening its patents related to fuel cell technology in an effort to, in the words of a Toyota executive, “speed the development of new technologies and move into the future of mobility more quickly, effectively and economically.”<sup>62</sup> Later that year, Ford similarly made its electrical vehicle patents available to EV developers, stating that doing so would “accelerate industry-wide research and development of electrified vehicles.”<sup>63</sup>

Several major automakers had released or were planning to release electric vehicles by 2016: Mercedes first released an electric version of its B-class sedan in 2014. Volkswagen released an electric version of its popular Golf compact in 2015.<sup>64</sup> In January 2015, GM announced a compact EV called the Chevrolet Bolt aimed at the same market segment as Tesla's Model 3. Slated for release by the end of 2016, it had a range certified by the EPA at 238 miles and an estimated sale price of about \$37,500.<sup>65</sup> In 2015, Audi, Mercedes, and Porsche announced plans for electric luxury vehicles that would compete with the Model S and Model X.<sup>66</sup>

EV manufacturers faced unavoidable trade-offs between cost and range of batteries. Tesla had, with the Model S and Model X, chosen to maximize range by equipping them with a large battery: a 70 kWh or 85 kWh battery delivered an EPA-estimated range of 240 to 265 miles, respectively, on a single charge. In 2016, Tesla announced the availability of 100 kWh battery packs for both vehicles, pushing the estimated range of the Model S over 300 miles.<sup>67</sup> But that range came at a cost: the size of the battery contributed heavily to the vehicles' price, which could be as high as \$135,000. Other manufacturers had opted to sacrifice range to cut costs. Nissan's Leaf, for example, originally came with a 24-kWh battery that gave the car a range of 84 miles per charge, while BMW's i3 had a 22-kWh battery that delivered a range of 81 miles.<sup>68</sup> Offerings from Volkswagen and Mercedes similarly had smaller batteries with ranges between 70 and 90 miles.

Tesla and other automakers partnered with major technology companies, including NEC, LG Chem, Samsung SDI, and Panasonic, to supply battery technology for their vehicles. Early in its history, Tesla had made the choice to construct its battery from existing industrial-grade li-ion cells manufactured by Panasonic, rather than build or outsource a battery specially designed for its vehicle, as other EV manufacturers had done. The form factor cells used by Tesla were slightly larger than an AA battery and were commonly used in consumer electronics. Tesla worked with Panasonic to optimize the cells for its vehicles.

## Energy Storage

In addition to electric vehicles, energy storage for residential, business, and utility use was a major potential market for battery technology. The widespread adoption of renewable energy sources such as wind and solar would require significant increases in storage capacity. Falling renewable energy costs (the cost of solar panels had fallen over 85% between 2007 and 2014) had led to increased uptake around the world. By 2015, the world was adding more renewable generating capacity than that of gas, coal, and oil combined. In many European countries, wind and solar accounted for over 20% of electricity generation, and Germany had set a target of getting 80% of its energy from renewables by 2050.<sup>69</sup> To reach such goals, renewable energy generation would have to be paired with large-scale storage capacity, most likely in the form of batteries, allowing energy generated by solar panels or wind turbines to be stored for use when the sun was not shining or the wind was not blowing.

As well as facilitating renewable energy use, load-balancing for electric utilities was another potential application; massive batteries could permit utility companies to avoid the necessity of firing



up so-called “peaker” power plants to meet peak demand. Instead, utilities could draw power from batteries at times of peak demand and recharge them when demand was low. As Musk put it in 2015, “you can basically, in principle, shut down half of the world’s power plants if you had stationary storage.”<sup>70</sup> Tesla saw this as an important secondary market for its batteries; in fact, Musk estimated in 2015 that the long-term capacity demand for stationary energy storage would be roughly double the demand for batteries for electric vehicles.<sup>71</sup> Indeed, in September 2016, Southern California Edison announced it would purchase 20 megawatts of Tesla batteries to plug into one of its substations, charging them up during times of low usage and discharging them at times of high demand.<sup>72</sup>

To take advantage of the energy storage market, Tesla announced in early 2015 its Tesla Energy suite of battery packs, designed to provide stationary storage for residential, commercial, and utility-scale applications. The residential product, called Powerwall, was designed for load-shifting (charging when rates were lower and discharging when rates were higher), to provide backup power during outages, and to be used in conjunction with rooftop solar applications, storing surplus solar energy for use when the sun was not shining. The Powerwall battery pack provided 6.4 kWh of energy storage, at a price of approximately \$3,000. Multiple battery packs could be installed together for greater capacity if needed. In October 2016, Tesla unveiled the Powerwall 2, along with solar roof-top panels that integrated directly with the energy storage system, at an event in Los Angeles. The second edition was priced higher at \$5,500.

Although the residential system garnered the most attention, Tesla executives anticipated that commercial and utility-scale applications would be much larger; Musk estimated that 5–10 times more capacity would be deployed at industrial and utility scale than at the consumer scale. A few months after the launch of Tesla Energy, Tesla CTO Straubel said that about 70% of reservations had been for the industrial-scale Powerpack and 30% for the residential-scale Powerwall.<sup>73</sup> Musk claimed the demand for Tesla’s energy storage solutions had been “staggering” in the first few months after announcing the product. By the end of the second quarter of 2015, Tesla reported that it had received more than 100,000 battery reservations, worth \$1 billion if they turned into sales (which Musk admitted they might not).<sup>74</sup> Musk insisted that, in order to realize the goal of transitioning the world’s electricity generation to renewable sources, Tesla hoped other battery companies would join Tesla in building “Gigafactory class operations of their own.” He also said that Tesla would continue its policy of “open-sourcing” its information technology related to the Gigafactory and battery manufacturing.<sup>75</sup> The opportunity to manufacture batteries for various energy storage solutions, whether targeted for electric vehicles or renewable energy, demonstrated such great potential to Elon Musk that he made the bold move to further vertically integrate Tesla by merging the electric vehicle and battery manufacturing company with SolarCity, an alternative energy sourcing and storing company.

## Tesla’s Renewable Energy Division: Solar City

Tesla and SolarCity had deeply linked histories and close ties. SolarCity’s founders, brothers Lyndon and Peter Rive, were Musk’s cousins; and it was Musk, by all accounts, who encouraged them to look into the solar energy business in 2004. Musk became chairman and a major financial backer when SolarCity was founded in 2006, and was its largest shareholder when it went public in 2012. He remained chairman and the largest shareholder in 2016. In addition, Tesla CTO Straubel sat on SolarCity’s board, and the firms shared one additional board member.<sup>76</sup>

For most of its history, SolarCity positioned itself as a solar energy provider; it purchased solar systems and installed them for residential, business, governments, and ultimately utility companies. It covered the cost of the installation and maintenance of the system and charged customers for the

energy produced by the system. Residential customers could pay as little as zero for the installation and pay a monthly fee for the electricity produced by the system, typically at a cost lower than that charged by the local utility.<sup>77</sup> The option to transition to solar power with no up-front costs had spurred adoption, and the number of SolarCity installations grew rapidly, with the installed customer base reaching 300,000 by the time Tesla acquired the company in November 2016.<sup>78</sup> Revenues had also increased significantly, from \$60 million in 2011 to \$538 million for the year ending June 30, 2016. Despite rapidly increasing revenues, SolarCity's debt levels grew at a much quicker pace (see **Exhibit 6** for SolarCity financial data).

Since SolarCity had not historically manufactured solar systems, it had benefited from the falling price of solar panels, which had enabled the cost of its rooftop installations to decline from \$4.73 in 2012 to \$2.84 per watt of generating power in 2016. In 2014, however, SolarCity acquired panel maker Silevo. The acquisition made SolarCity, in the words of Peter Rive, "the most vertically integrated solar company in the world."<sup>79</sup> To that end, SolarCity began construction on a \$750 million factory in Buffalo, New York, where it would manufacture solar panels in 2017. SolarCity officials claimed that the new panels would be as much as 30% more efficient than the commodity solar panels it was currently purchasing. These efficiency gains, combined with a simplified manufacturing process and the scale of production, had the potential of even further reducing the cost of residential solar installations to \$2.50/watt.<sup>80</sup>

Declining costs for producing solar translated into greater competition for SolarCity. The residential and commercial markets for solar installation were both highly competitive. According to one analyst, the top 10 residential solar installers in 2015 represented 58% of the market, with SolarCity leading at 35% (Vivint Solar was second at 11%). On the other hand, the top 10 producers in the commercial market represented only 42%, with SolarCity leading at 14% (SunPower was second at 7%). "The fragmented commercial developer landscape is largely the result of bottlenecks in the customer origination process that make it difficult for any individual player to consistently grow," said noted one research analyst. SolarCity would need to defend its lead position among two different sets of competitors, while at the same time, barriers to entry were falling.<sup>81</sup> To do so required high spending; SolarCity's sales, administrative, and research costs were \$438 million in the first half of 2016, 42% greater than revenue of \$308 million.<sup>82</sup> Such a high level of spending meant that SolarCity needed to find a solution to its continuing need for financing.

Further indicating its commitment to the renewable energy business, Tesla first announced in June 2016 a proposal to acquire SolarCity. The acquisition offer was all-stock, in which each share of SolarCity would be exchanged for a fraction of a share in Tesla. This exchange ratio of stock represented a value of \$26.50-\$28.50 per share of SolarCity and a total equity value of \$2.6-\$2.8 billion. Tesla's shareholders did not react positively to the announcement; the next day Tesla's stock price fell by 10.5%, while SolarCity's stock price rose 3.3%.<sup>83</sup> In the context of the SolarCity acquisition, Musk pointed out that renewable energy generation had long been a part of Tesla's vision. "The point of all this," he wrote in July 2016, "was, and remains, accelerating the advent of sustainable energy." Indeed, the original "master plan" written by Musk in 2006, though focused on Tesla's EV business, included the goal of becoming "energy positive" by providing "zero emission electric power generation options," partly through partnership with the then newly founded SolarCity.<sup>84</sup>

SolarCity had begun offering home energy bundles including solar panels and battery backup using Tesla's Powerwall battery packs earlier in the year, and for Musk the merger was a natural progression. Musk argued that there was considerable overlap between the two companies' potential customer bases, that Tesla's design and manufacturing experience would benefit SolarCity's solar panels, while SolarCity's sales, distribution, and installation capabilities could serve Tesla customers.<sup>85</sup> Tesla claimed \$150 million in cost synergies that would be unlocked in the merger (see **Exhibit 7**). JP Morgan's research

report on the initial offer cast doubt on the customer overlap: "Absent a detailed explanation (at this time) we are struggling to see brand, customer, channel, product or technology synergies."<sup>86</sup> (See **Exhibits 8 and 9**.) Several analysts criticized the proposed deal as adding to Tesla's already daunting challenges of scaling battery and vehicle production, but Musk insisted that Tesla would benefit from creating a "smoothly integrated and beautiful solar-roof-with-battery product. We can't do this well if Tesla and SolarCity are different companies, which is why we need to combine and break down the barriers inherent to being separate companies."<sup>87</sup> In anticipation of the completion of the merger, Musk announced a new solar roof product which replaced a traditional roof with glass tiles that contained power-generating solar cells. Musk later claimed that Tesla's solar roofing material would cost less than traditional premium roofing materials, even before factoring in savings from electricity generation.<sup>88</sup>

SolarCity's financial struggles also contributed to skepticism about the proposed acquisition. The company struggled to become profitable and its operating losses were mounting – totaling nearly \$980 million for 2014–2015, leading to considerable doubts about merging two companies that were steadily burning through cash, to the tune of an estimated \$2.5 billion combined in 2016.<sup>89</sup> (See **Exhibits 3b and 6b** for Tesla and SolarCity cash flow data.) The *Wall Street Journal* wrote that "Tesla latching on to SolarCity is the equivalent of a shipwrecked man clinging to a piece of driftwood grabbing on to another man without one," while *Forbes* wondered how the already-indebted Tesla could carry SolarCity's \$5 billion long-term debt load.<sup>90</sup> Some critics argued the acquisition was an effort to save SolarCity from its debt obligations, which grew 13x from 2013 to 2016. "This deal has everything to do with debt. Call it a bailout, call it what you will. . . . SolarCity is one bad economic downturn away from going belly up," said one analyst.<sup>91</sup> Musk and Tesla responded by pointing out that SolarCity's debt position and cash flows appeared to be worse than they really were because SolarCity used debt to finance rooftop installations that generated revenue over time through long-term leases and power purchase agreements.<sup>92</sup>

Tesla revised its offer for SolarCity on August 1, 2016, in part due to SolarCity announcing that its 2016 forecast for megawatts installed would be 10% lower than previously expected. The new acquisition offer was about \$300 million lower than the initial offer at \$25.83 per share, representing a \$2.6 billion equity valuation.<sup>93</sup> This price for SolarCity was 10x higher than the average revenue multiple of SolarCity's publicly traded peers (see **Exhibit 10**). On the other hand, \$25.83 per share was well below SolarCity's all-time-high closing price of \$86.14 set in February 2014 (see **Exhibit 11**).

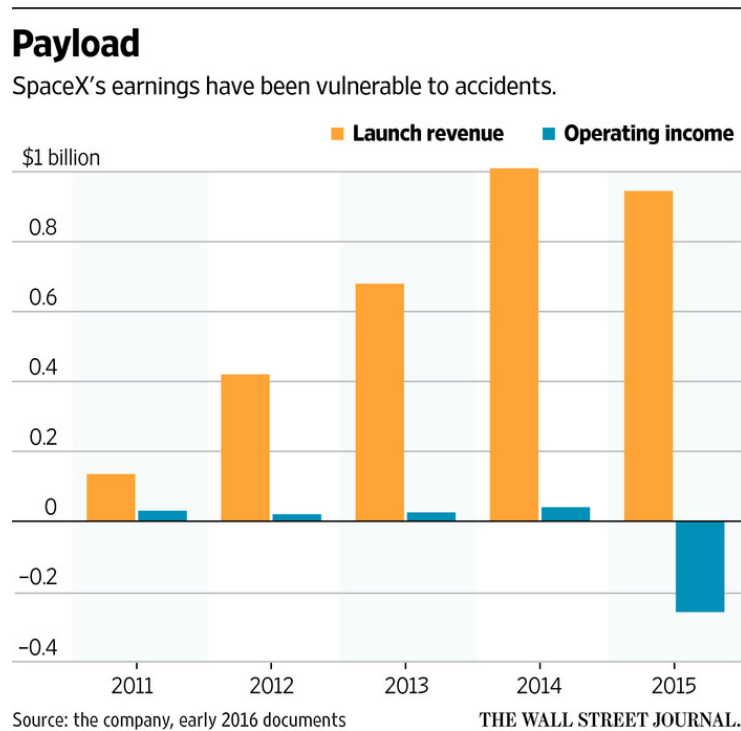
Corporate governance issues were also debated. The required number of shareholders from both companies to approve the merger was considerably lower than normal. JP Morgan's research remarked: "The proposal requires the majority approval . . . from shareholders accounting for 39.44% of economic interest in Tesla (50% of shareholders excluding Elon Musk's interest of 21.12%) and 38.73% of economic interest in SolarCity (50% of shareholders excluding Elon Musk's interest of 22.54%)."<sup>94</sup> Due to conflicts of interest, several board members of both companies had to abstain from voting on the merger. On the Tesla side, Elon Musk and Antonio Gracias recused themselves because they served on both boards. On the SolarCity side, five of eight board members recused themselves given their various connections to Elon Musk and Tesla. This left two independent board members, Donald Kendall and Nancy Pfund, who were solely responsible for approving the merger on behalf of the SolarCity board.<sup>95</sup>

In the end, investors sided with Musk against the skeptics. After the boards of both companies approved the proposed merger in August, shareholders overwhelmingly approved the acquisition in November, with over 85% of Tesla shareholders voting in favor of the deal.<sup>96</sup>

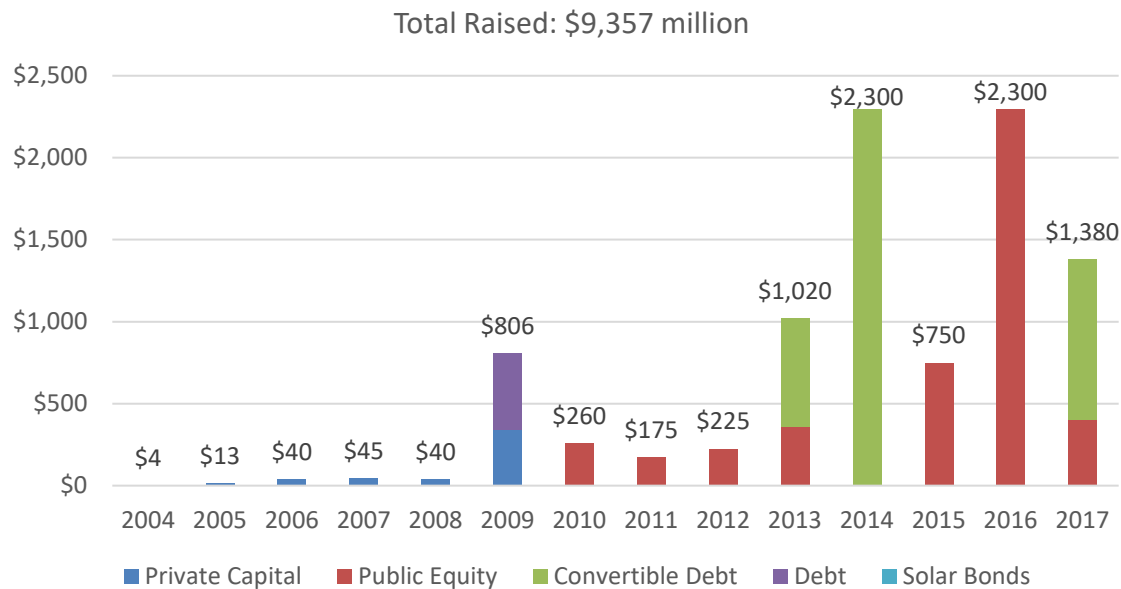
## Conclusion

Ramping vehicle production, completing the first Gigafactory, and growing SolarCity each required significant capital investments at a time when Tesla was still losing money. Musk and his companies owed some \$6 billion in debt to bondholders and lenders, including nearly \$500 million in debt held by Musk himself, who had taken out personal lines of credit over the years in order to purchase additional shares of Tesla and SolarCity when they needed infusions of capital. Musk's companies also lent to each other: SpaceX purchased \$165 million in bonds from SolarCity in 2014 and 2015.<sup>97</sup> (See **Exhibit 12**.) As the largest shareholder, Musk's holdings in Tesla/SolarCity and SpaceX were 21% and 54%, respectively. He had invested almost all of his personal fortune in his various ventures, and though his stake in the companies gave him a net worth of about \$13 billion by March 2017, he had little cash on hand. In 2016, SpaceX lost money and the combined Tesla/SolarCity reported a \$667 million operating loss (see **Exhibit 3a**).

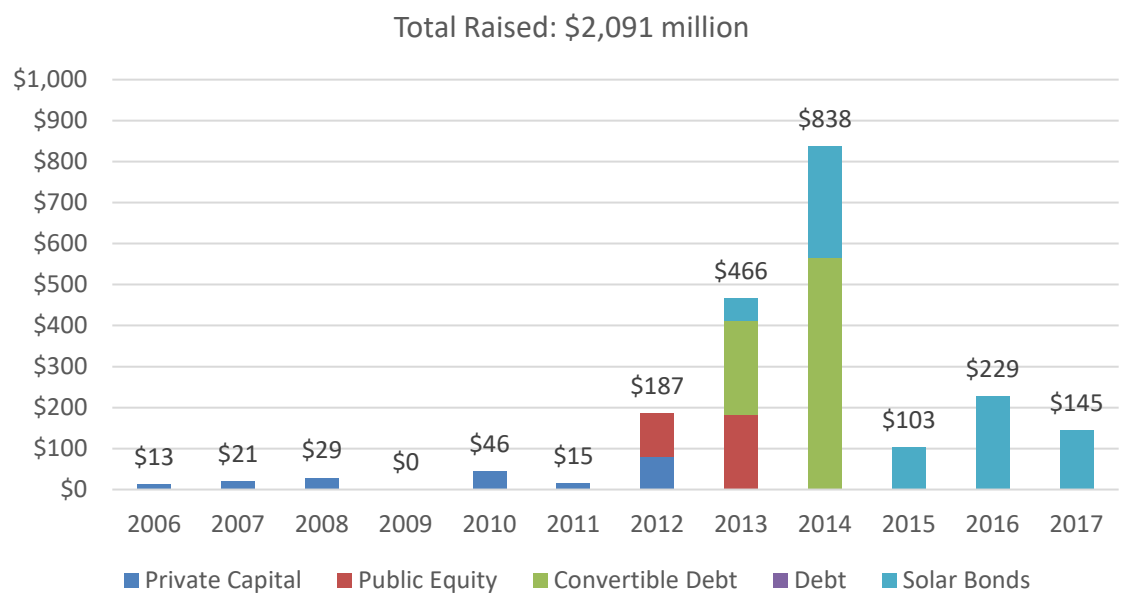
Despite the mounting losses, and despite Musk's repeated promises that he would not need new capital in 2017, Tesla successfully raised nearly \$1.4 billion in equity and convertible debt in March 2017 to continue moving forward with the Model 3 production.<sup>98</sup> Tesla also welcomed a new investor: Chinese giant Tencent spent \$1.8 billion to buy 5% of Tesla and become one of the company's largest shareholders.<sup>99</sup> Staying true to character, Elon Musk invested another \$25 million of his own money. With the expansion of the Gigafactory and the acquisition of SolarCity, observers wondered what direction Tesla would take. Was the end game for Musk to be a world leader in electric cars, with its battery manufacturing capacity serving primarily to help drive the adoption of its mass-market electric car? Or would Tesla become primarily a battery manufacturer, possibly supplying batteries for other electric vehicles as well as for the burgeoning stationary storage energy business? Or were his ambitions greater, using the proposed acquisition of SolarCity to accelerate the adoption of renewable energy worldwide? On the one hand, Musk's vision and ambitions were breathtaking: he had built three companies worth more than \$60 billion in April 2017. On the other hand, despite a lack of profits, Musk was asking Wall Street and shareholders to trust that he could pull together multiple businesses, and join him in making multiple big bets.

**Exhibit 1** SpaceX Annual Revenue and Operating Income (\$ millions)

Source: Rolfe Winkler and Andy Pasztor, "Exclusive Peek at SpaceX Data Shows Loss in 2015, Heavy Expectations for Nascent Internet Service," *Wall Street Journal*, January 13, 2017, <https://www.wsj.com/articles/exclusive-peek-at-spacex-data-shows-loss-in-2015-heavy-expectations-for-nascent-internet-service-1484316455>, accessed February 27, 2017.

**Exhibit 2a** Tesla's Fundraising History (\$ millions) through March 2017

Source: Created by casewriter using Tesla's 424B4 and 424B5 public filings on [www.SEC.gov](http://www.SEC.gov), Thomson One Banker, and Pitchbook, accessed March 2017.

**Exhibit 2b** SolarCity's Fundraising History (\$ millions) through March 2017

Source: Created by casewriter using SolarCity's 424B4 and 424B5 public filings on [www.SEC.gov](http://www.SEC.gov), Thomson One Banker, and Pitchbook, accessed March 2017.

**Exhibit 3a** Tesla Selected Income and Balance Sheet Data, 2011–2016 (\$ millions)

|                                 | 2011     | 2012     | 2013     | 2014     | 2015      | 2016*    |
|---------------------------------|----------|----------|----------|----------|-----------|----------|
| Revenues                        | 204.2    | 413.3    | 2,013.5  | 3,198.4  | 4,046.0   | 7,000.1  |
| Cost of sales                   | 142.7    | 383.2    | 1,557.2  | 2,316.7  | 3,122.5   | 5,400.9  |
| R&D                             | 209.0    | 274.0    | 232.0    | 464.7    | 717.9     | 834.4    |
| SG&A                            | 104.1    | 150.4    | 285.6    | 603.7    | 922.2     | 1,390.3  |
| Operating income (loss)         | (251.5)  | (394.3)  | (61.3)   | (186.7)  | (716.6)   | (667.3)  |
| Net income (loss)               | (254.4)  | (396.2)  | (74.0)   | (294.0)  | (888.7)   | (773.0)  |
| EBITDA                          | (234.6)  | (365.5)  | 44.8     | 45.2     | (294.0)   | 319.9    |
| Cash and ST investments         | 280.3    | 201.9    | 845.9    | 1,905.7  | 1,196.9   | 3,393.2  |
| Total assets                    | 713.4    | 1,114.2  | 2,416.9  | 5,830.7  | 8,067.9   | 22,664.1 |
| Total debt                      | 280.1    | 466.7    | 606.9    | 2,519.5  | 2,897.7   | 8,588.1  |
| Net debt                        | (0.2)    | 264.8    | (239.0)  | (613.8)  | (1,700.8) | 5,194.9  |
| Total liabilities               | 489.4    | 989.5    | 1,749.8  | 4,919.0  | 6,984.2   | 16,750.1 |
| Total shareholders' equity      | 224.0    | 124.7    | 667.1    | 911.7    | 1,083.7   | 4,752.9  |
| Total debt/equity               | 125.0%   | 374.2%   | 91.0%    | 276.3%   | 267.4%    | 145.4%   |
| Gross margin                    | 30.2%    | 7.3%     | 22.7%    | 27.6%    | 22.8%     | 22.8%    |
| R&D/sales                       | 102.4%   | 66.3%    | 11.5%    | 14.5%    | 17.7%     | 11.9%    |
| Return on equity                | (118.0%) | (227.2%) | (18.7%)  | (37.2%)  | (89.1%)   | (22.1%)  |
| Market value (at annual filing) | 3,506.5  | 4,349.8  | 31,142.0 | 25,977.9 | 23,638.1  | 40,221.7 |

Source: Created by casewriter using data from Capital IQ and Tesla's 10-K filings, accessed March 2017.

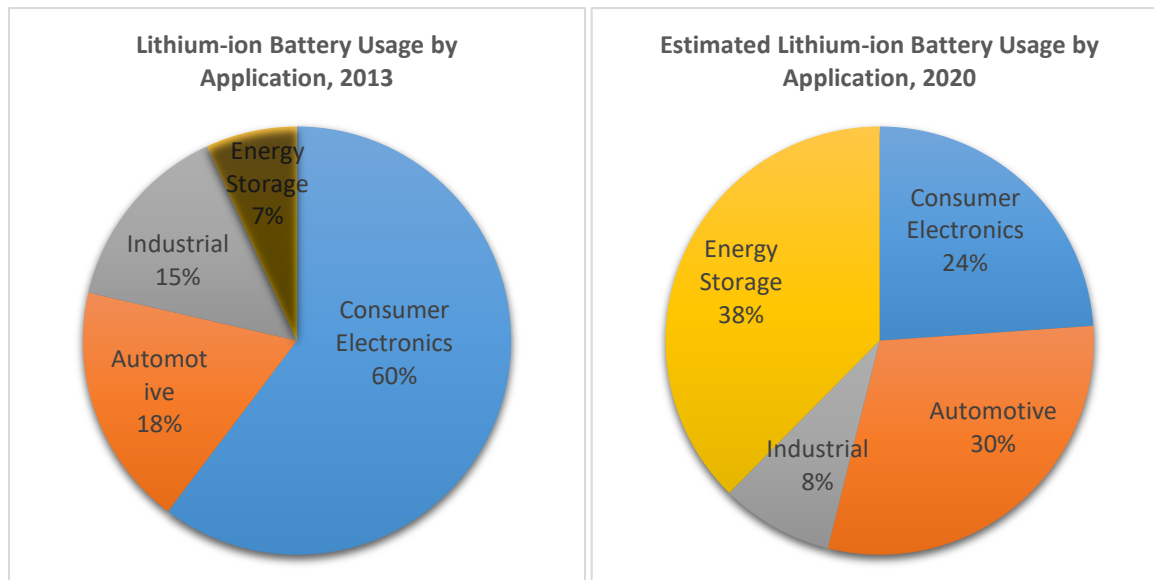
\* 2016 financials include SolarCity from the date of acquisition: November 21, 2016–December 31, 2016.

**Exhibit 3b** Tesla Selected Cash Flow Data (\$ millions)

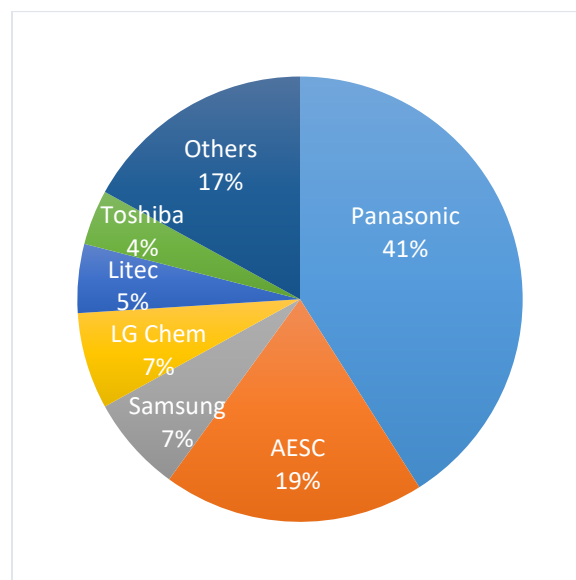
|                                         | 2011    | 2012    | 2013    | 2014    | 2015      | 2016*     |
|-----------------------------------------|---------|---------|---------|---------|-----------|-----------|
| Net cash flow from operating activities | (128.0) | (263.8) | 264.8   | (57.3)  | (524.5)   | (123.8)   |
| Net cash flow from investing activities | (162.3) | (206.9) | (249.4) | (990.4) | (1,673.6) | (1,416.4) |
| Net cash flow from financing activities | 446.0   | 419.6   | 635.4   | 2,143.1 | 1,523.5   | 3,744.0   |
| Net increase in cash                    | 155.7   | (53.4)  | 644.0   | 1,059.8 | (708.8)   | 2,196.3   |
| Cash at beginning of period             | 99.6    | 255.3   | 201.9   | 845.9   | 1,905.7   | 1,196.9   |
| Cash at end of period                   | 255.3   | 201.9   | 845.9   | 1,905.7 | 1,196.9   | 3,393.2   |

Source: Created by casewriter using data from Capital IQ and Tesla's 10-K filings, accessed March 2017.

\* 2016 financials include SolarCity from the date of acquisition: November 21, 2016–December 31, 2016.

**Exhibit 4** Lithium-ion Battery Usage, 2013–2020

Source: Adapted from Vishal Sapru, "Analysis of the Global Lithium-ion Battery Market: Growth Opportunities and Market Outlook," Frost & Sullivan Market Research, June 10, 2014.

**Exhibit 5** Lithium-ion Battery Manufacturers Market Share, 2015

Source: Adapted from "Lithium-ion Batteries—Statista Dossier," Statista, Inc., 2015.



**Exhibit 6a** SolarCity Selected Financial Data (\$ millions)

|                                 | 2011    | 2012    | 2013    | 2014    | 2015    | 2016    |
|---------------------------------|---------|---------|---------|---------|---------|---------|
| Revenues                        | 59.6    | 126.9   | 163.8   | 255.0   | 399.6   | 730.3   |
| Cost of sales                   | 47.1    | 99.5    | 124.5   | 176.4   | 280.8   | 478.9   |
| R&D                             | 0.0     | 0.0     | 1.5     | 19.2    | 64.9    | 55.0    |
| SG&A                            | 73.7    | 118.5   | 185.8   | 394.7   | 701.7   | 755.6   |
| Operating income (loss)         | (61.3)  | (91.0)  | (148.0) | (335.3) | (647.8) | (650.3) |
| EBITDA                          | (48.9)  | (70.2)  | (106.5) | (237.4) | (481.1) | (375.7) |
| Cash and ST investments         | 50.5    | 160.1   | 577.1   | 642.7   | 393.9   | 290.7   |
| Total assets                    | 813.2   | 1,342.3 | 2,809.5 | 4,551.2 | 7287.1  | 9,130.8 |
| Total debt                      | 98.1    | 289.5   | 665.6   | 1,516.6 | 2,867.5 | 3,687.5 |
| Net debt                        | 47.6    | 129.4   | 88.5    | 873.9   | 2,473.6 | 3,396.8 |
| Total liabilities               | 597.1   | 1,049.1 | 1,960.4 | 3,208.8 | 5,552.6 | 6,855.7 |
| Total shareholders' equity      | 216.0   | 293.2   | 849.1   | 1,342.4 | 1,734.6 | 2,275.1 |
| Total debt/equity               | 45.4%   | 98.7%   | 78.4%   | 113.0%  | 165.3%  | 162.1%  |
| Gross margin                    | 20.8%   | 21.6%   | 24.0%   | 30.8%   | 29.7%   | 24.9%   |
| R&D/sales                       | 0.0%    | 0.0%    | 0.9%    | 7.5%    | 16.2%   | 7.5%    |
| SG&A/sales                      | 123.7%  | 93.3%   | 113.4%  | 154.8%  | 175.6%  | 103.5%  |
| Return on equity                | (40.9%) | (44.7%) | (26.6%) | (34.2%) | (50.0%) | (40.9%) |
| Market value (at annual filing) | n/a     | 1,422.0 | 7,055.0 | 5,124.2 | 1,824.1 | N/A     |

Source: Created by casewriter using data from Capital IQ and SolarCity's 10-K filings, accessed March 2017.

**Exhibit 6b** SolarCity Selected Cash Flow Data (\$ millions)

|                                         | 2011    | 2012    | 2013    | 2014      | 2015      | 2016      |
|-----------------------------------------|---------|---------|---------|-----------|-----------|-----------|
| Net cash flow from operating activities | 16.8    | 39.8    | 174.5   | (217.8)   | (789.9)   | (508.8)   |
| Net cash flow from investing activities | (302.9) | (428.5) | (729.9) | (1,344.8) | (1,726.7) | (1,689.3) |
| Net cash flow from financing activities | 278.4   | 498.3   | 972.4   | 1,490.0   | 2,394.8   | 2,106.3   |
| Net increase (decrease) in cash         | (7.8)   | 109.6   | 417.0   | (72.7)    | (121.8)   | (91.8)    |
| Cash at beginning of period             | 58.3    | 50.5    | 160.1   | 577.1     | 504.4     | 382.5     |
| Cash at end of period                   | 50.5    | 160.1   | 577.1   | 504.4     | 382.5     | 290.7     |

Source: Created by casewriter using data from Capital IQ and SolarCity's 10-K filings, accessed March 2017.

**Exhibit 7** Tesla's Presented Synergies of SolarCity Acquisition

## Cost Synergies

- \$150mm of direct cost synergies expected to be achieved in first full year after closing
- Cost synergies driven by:
  - Sales and marketing efficiencies enabled through cross-selling
  - Significant corporate and overhead savings
- Additional potential cost savings:
  - Lowering hardware costs and total cost of ownership (TCO)
  - Reducing installation and service costs
  - Improving manufacturing efficiency
  - Reducing customer acquisition costs through combined sales channels
  - Optimizing capital expenditure costs

## Revenue Synergies

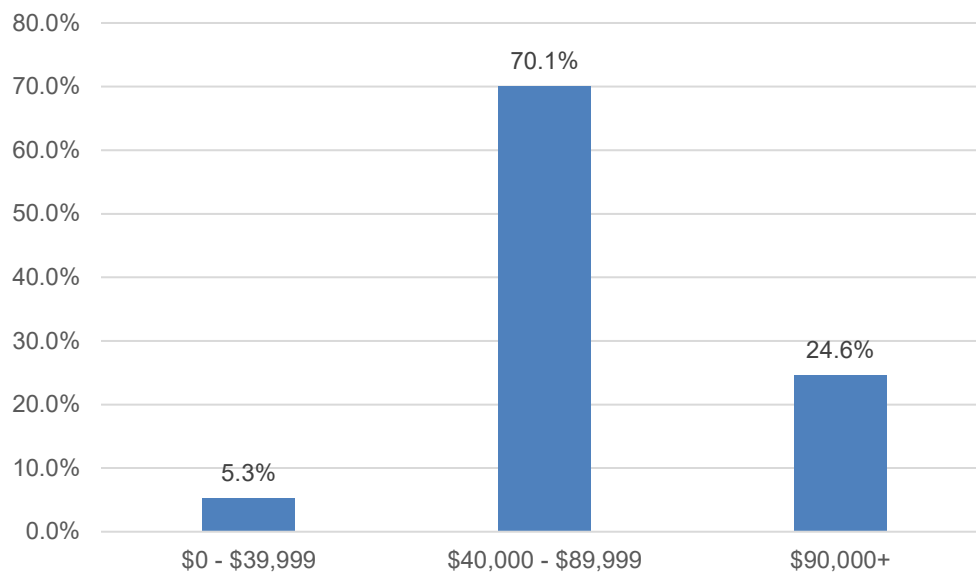
- Leverage Tesla's footprint to drive solar sales leads and consumer trust
- Increase solar leads from foot traffic in Tesla stores – customers have overlapping product interest
- Global reach of Tesla's 190+ stores and brand as solar product expands internationally
- Simplifying the install process for vehicle charging, solar, and storage products
- Single ordering experience, installation, and service contract

Source: Tesla blog post, "Tesla and SolarCity," November 1, 2016, <https://www.tesla.com/blog/tesla-and-solarcity>, accessed February 27, 2017.

**Exhibit 8** Tesla Model S Buyer Profile

|                           |       |
|---------------------------|-------|
| % Male:                   | 83.9% |
| % Female:                 | 16.1% |
| Income under \$50,000:    | 5.7%  |
| Income \$50,000–\$99,999: | 17.2% |
| Income over \$100,000:    | 77.3% |
| 18–44 yrs. Old:           | 33.2% |
| 45–64 yrs. Old:           | 50.6% |
| 65+ yrs. Old:             | 16.2% |

Source: Created by casewriter using data from: Phil LeBeau, "The Young and Rich Snapping Up Tesla's Model S," CNBC, August 19, 2013, <http://www.cnbc.com/id/100970641>, accessed February 27, 2017. Data from: Edmunds.com.

**Exhibit 9** Income Distribution among Rooftop Solar Installations in California, Arizona, and New Jersey

Source: Mari Hernandez, "Solar Power to the People: The Rise of Rooftop Solar Among the Middle Class," Center for American Progress, October 21, 2013, <https://www.americanprogress.org/issues/green/reports/2013/10/21/76013/solar-power-to-the-people-the-rise-of-rooftop-solar-among-the-middle-class/>, accessed February 27, 2017. Distribution averaged across data sets for all three states: California, New Jersey, and Arizona.

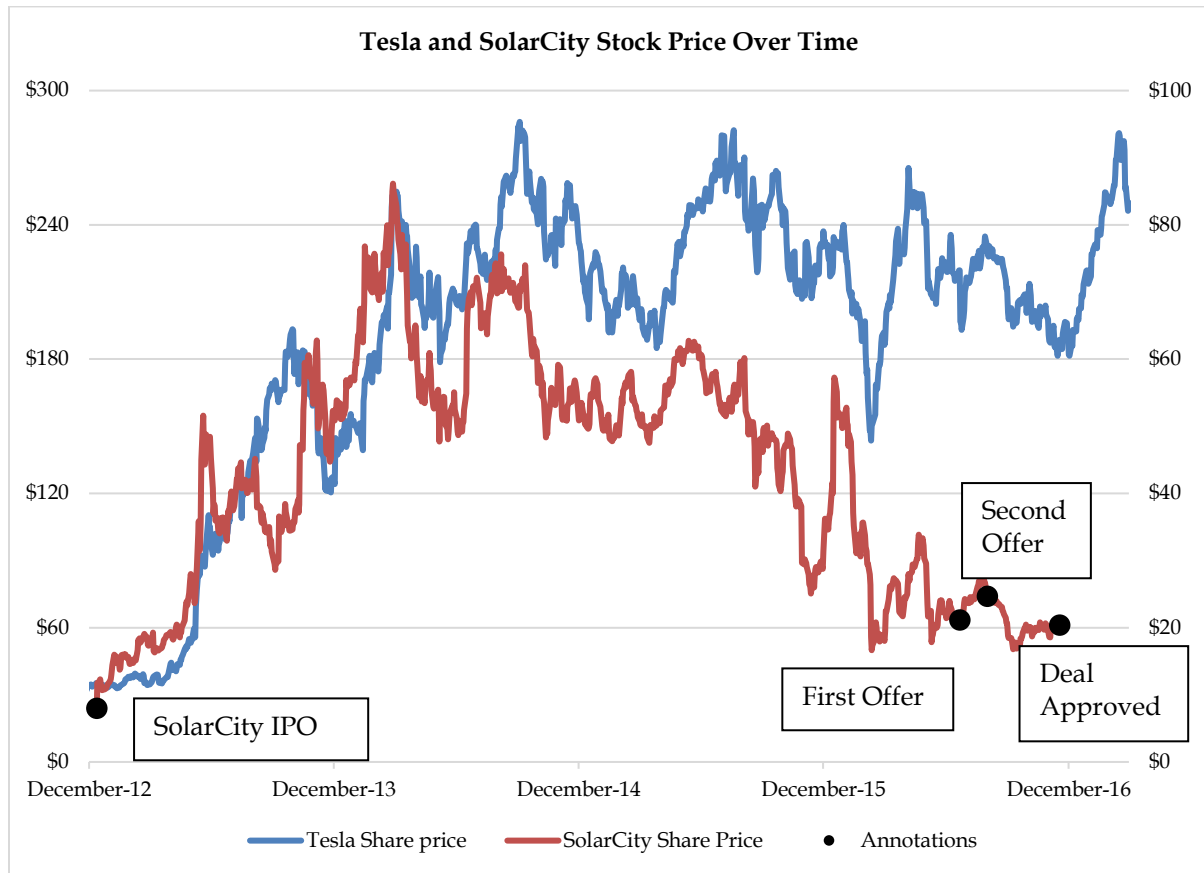
**Exhibit 10** SolarCity Acquisition Price Compared to Peers (\$ millions, prices as of November 21, 2016—closing date of SolarCity acquisition)

| Company                 | Market Value   | Net Debt & Minority Interest | Enterprise Value | Enterprise Value / LTM Revenue | Enterprise Value / NTM Revenue |
|-------------------------|----------------|------------------------------|------------------|--------------------------------|--------------------------------|
| <b>SolarCity</b>        | <b>\$2,687</b> | <b>\$4,235</b>               | <b>\$6,921</b>   | <b>12.87x</b>                  | <b>10.63x</b>                  |
| <b>Average of Peers</b> | <b>\$832</b>   | <b>\$1,080</b>               | <b>\$1,922</b>   | <b>2.47x</b>                   | <b>1.77x</b>                   |
| First Solar             | \$3,053        | (\$1,264)                    | \$1,789          | 0.52x                          | 0.82x                          |
| Trina Solar             | \$961          | \$1,471                      | \$2,432          | 0.70x                          | 0.74x                          |
| SunPower                | \$915          | \$2,065                      | \$2,981          | 1.56x                          | 1.11x                          |
| Canadian Solar          | \$617          | \$2,013                      | \$2,630          | 0.80x                          | 0.91x                          |
| Sunrun                  | \$500          | \$1,151                      | \$1,651          | 3.80x                          | 3.00x                          |
| Vivint Solar            | \$297          | \$850                        | \$1,147          | 10.50x                         | 5.47x                          |
| JA Solar                | \$251          | \$594                        | \$864            | 0.37x                          | 0.42x                          |
| Yingli Green Energy     | \$61           | \$1,760                      | \$1,884          | 1.54x                          | 1.71x                          |

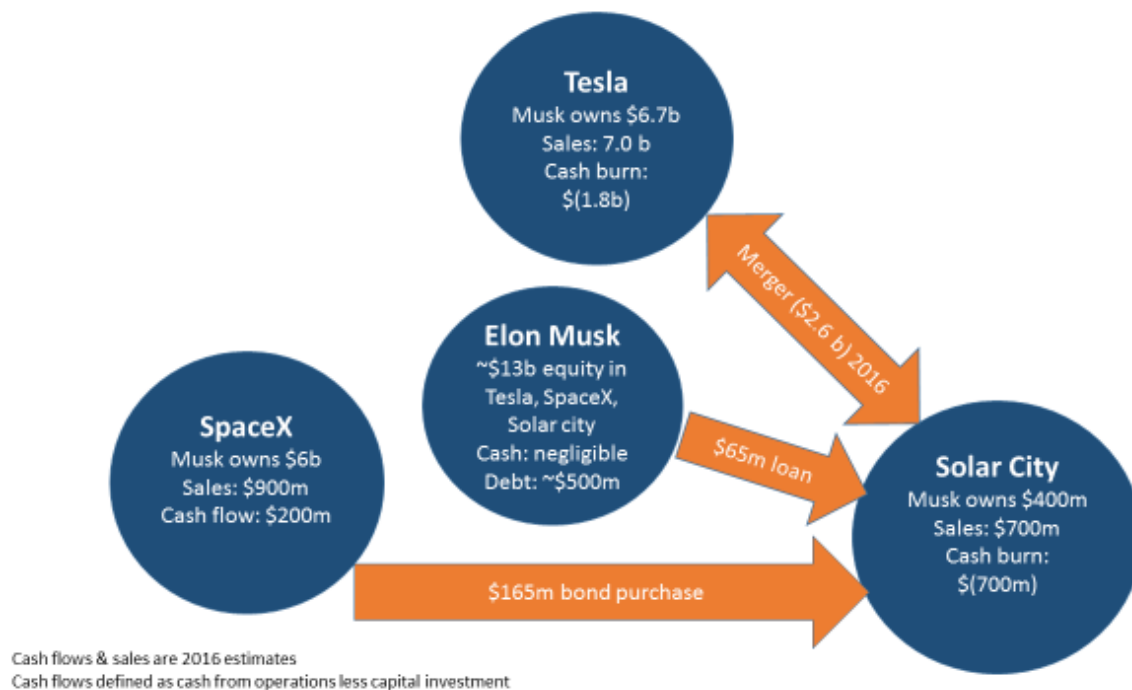
Definition of Terms:

- *Market Value = Total Shares Outstanding × Price Per Share*
- *Net Debt = Total Debt – Cash & Short Term Investments*
- *Enterprise Value = Market Value + Net Debt + Minority Interest*

Source: Created by casewriter using data from Capital IQ, accessed March 2017. Revenue forecasts reflect consensus estimates on November 21, 2016, the date of the SolarCity acquisition closing.

**Exhibit 11** Tesla and SolarCity Stock Price Over Time

Source: Created by casewriter using data from Capital IQ, accessed March 2017.

**Exhibit 12** Elon Musk's Empire

Source: Adapted from "Countdown: Elon Musk's Empire," *The Economist*, October 22, 2016, <http://www.economist.com/news/business/21709061-entrepreneurs-finances-are-jaw-dropping-inventive-and-combustible-his-space?frsc=dg%7Ca>, accessed October 30, 2016.

## Endnotes

- <sup>1</sup> Cindy Perman, "Tesla Beats Expectations, Delivering a Record 25,000 vehicles in Q1," CNBC, April 2, 2017, <http://www.cnbc.com/2017/04/02/tesla-delivers-a-record-25000-vehicles-in-the-first-quarter.html>, accessed April 11, 2017.
- <sup>2</sup> Noel Randewich, "Tesla Becomes Most Valuable U.S. Car Maker, Edges Out GM," Reuters, April 11, 2017, <http://www.reuters.com/article/us-usa-stocks-tesla-idUSKBN17C1XF>, accessed April 11, 2017.
- <sup>3</sup> Elon Musk, Twitter Account, April 3, 2017, <https://twitter.com/elonmusk/status/848935705057280001?lang=en>, accessed April 11, 2017.
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